



Forensic identification of fatal mushroom poisoning: a case report

Zhonghao Yu¹ · Yuhao Yuan¹ · Jiaxin Zhang¹ · Qing Shi¹ · Hang Gao² · Yiwu Zhou¹

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Abstract

Mushroom poisoning incidents happen infrequently, yet owing to the non-lethal nature of most toxins and the efficacy of timely treatment, fatalities from mushroom poisoning are uncommon, leading to a scarcity of pertinent clinical and pathological data. Here, we reported a case of death caused by the consumption of raw mushrooms, alongside detailed clinical data and multi-organs pathological alterations, which underscored its potential significant reference value in forensic practice. Furthermore, *ibotenic acid*, a type of mushroom toxin, was detected both in the patient's blood and gastric lavage fluid about 19 h after the consumption of mushrooms, and was successfully quantified at concentrations of 0.381 µg/mL and 0.202 µg/mL, respectively, indicating a relatively significant finding.

Keywords Forensic pathology · Mushroom poisoning · Ibotenic acid · Multi-organ failure

Introduction

Mushroom poisoning occurs frequently on a global scale. It was reported that a total of 4,412 people were hospitalized due to mushroom poisoning in Germany from 2000 to 2018, among which 22 people died [1], and a retrospective cohort study in Thailand also reported a total of 41 cases of myotoxic mushroom poisoning from 2012 to 2016, including 11

deaths [2]. Moreover, in 2023, the China CDC reported a total of 1,303 mushroom poisoning cases and 16 associated deaths [3]. Recently, with the increasing interest in consuming wild mushrooms and the globalization of food practices, the incidence of mushroom poisoning has been on the rise. This trend posed significant challenges in forensic pathology due to the diverse clinical presentations of poisoning and its potential for severe outcomes. Therefore, there is an urgent need for a more in-depth understanding of the forensic implications of mushroom poisoning.

Toxic mushrooms can be classified into six groups based on their characteristic and clinical manifestations: (1) cytotoxic; (2) neurotoxic; (3) myotoxic; (4) metabolic, endocrine, and related toxicity; (5) gastrointestinal irritant; and (6) miscellaneous adverse reactions to mushrooms [4]. However, the forensic pathology perspective regarding toxic mushrooms remains underexplored. In particular, there was a lack of in-depth research on the specific pathological findings and their implications in legal contexts. Accurate diagnosis of mushroom poisoning often required a detailed analysis of post-mortem tissue samples, which can be challenging due to the varied nature of mushroom toxins and the requirement for specialized testing.

Therefore, to better illustrate the characteristics of mushroom poisoning at various stages—both before and after death—we would like to present a detailed forensic analysis of a mushroom poisoning case to fill this knowledge gap by offering insights into the clinical and

Zhonghao Yu and Yuhao Yuan contributed equally to this work.

✉ Hang Gao
274843208@qq.com

✉ Yiwu Zhou
zhouyiwu@hust.edu.cn

Zhonghao Yu
1259970467@qq.com

Yuhao Yuan
362736952@qq.com

Jiaxin Zhang
296034673@qq.com

Qing Shi
806496165@qq.com

¹ Department of Forensic Medicine, Tongji Medical College, Huazhong University of Science and Technology, Wuhan 430030, China

² Huangshi Public Security Bureau, Huangshi 435000, China

pathological features observed. Through the documentation of this case, we sought to enhance the forensic community's understanding of mushroom poisoning, improve diagnostic protocols, increase diagnostic accuracy, and ultimately refine the management of such cases in legal contexts.

Case report

Case history

A 28-year-old man (height: 172 cm; weight: 80 kg) presented with abdominal pain and bloating just 30 min and became poisoned after consuming the raw mushroom slices that were placed as garnish beside a plate of sashimi in a restaurant. Over the subsequent 9 h, his condition deteriorated, developing symptoms such as difficulty breathing, hypotension,

disturbance of consciousness, and eventually cardiac arrest. Despite receiving intensive treatments, including venoarterial extracorporeal membrane oxygenation (VA-ECMO), blood transfusion, and correction of electrolyte imbalances, the patient unfortunately succumbed to the poisoning 76 h after ingesting the mushrooms. A detailed timeline of his symptoms and treatments during hospitalization was provided in Supplementary Table S1.

Medical records

Approximately 14 h after consuming the mushrooms, the man underwent VA-ECMO-based treatments, and various physical indicators were closely monitored throughout his hospitalization. As depicted in Fig. 1, despite the administration of treatment, the patient's blood gas analysis results remained relatively disturbed, indicating ongoing acid-base imbalances within the body. Furthermore, the abnormal

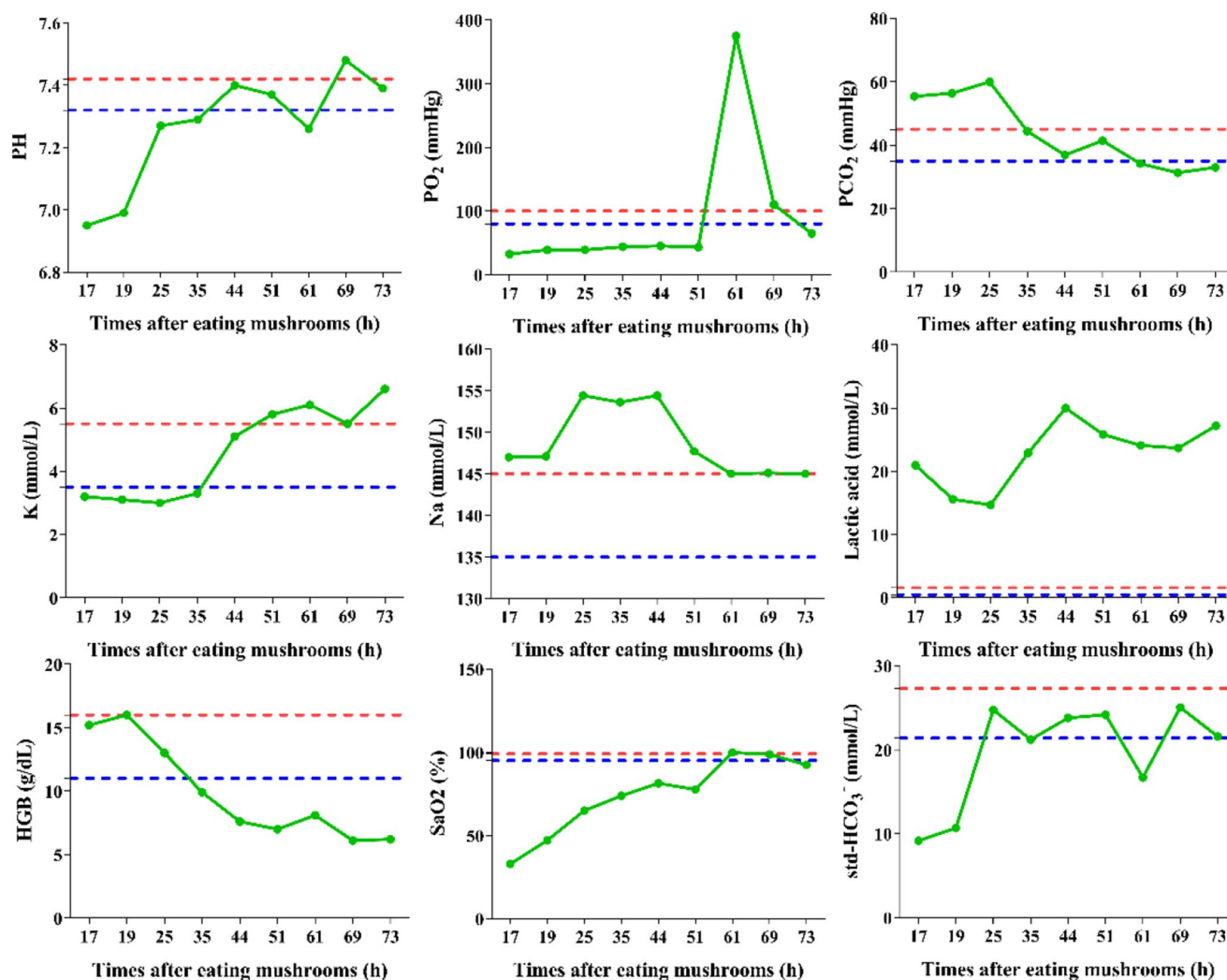


Fig. 1 Representative results of blood gas analysis during the hospitalization after eating mushrooms. (Note The red and blue dashed lines represent the upper and lower limits of the normal range)

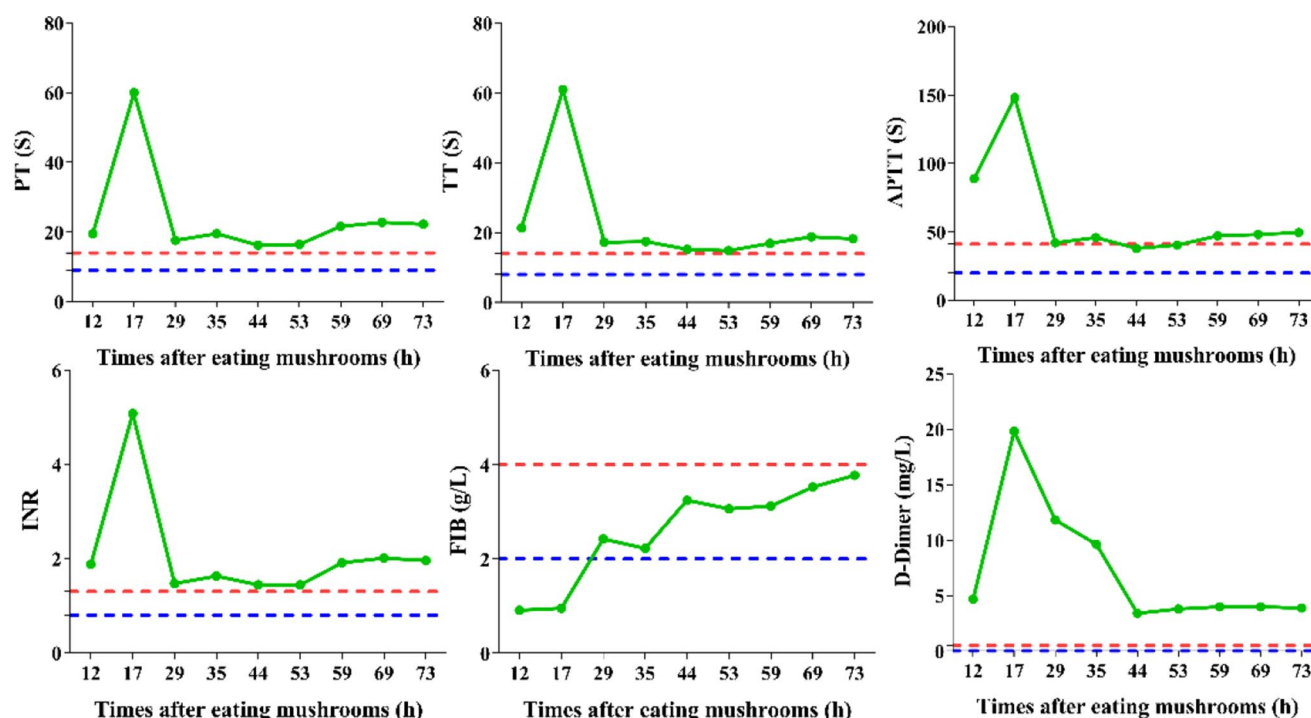


Fig. 2 Representative results of blood clotting test during the hospitalization after eating mushrooms. (Note The red and blue dashed lines represent the upper and lower limits of the normal range)

Table 1 Results of representative indicators of pancreas and myocardium during the hospitalization after eating mushrooms

Indicators of pancreas	Times after eating mushrooms		
	12h	35h	59h
Amylase (35~135 U/L)	815	1812	1385
Lipase (0~60 U/L)	163.4	1492	984
Indicators of myocardium	12h	26h	59h
LDH (120~250 U/L)	1396.8	2178	3788
HBDH (72~182 U/L)	\	1122	701
CK (50~310 U/L)	138.3	4892	6344
CK-MB (0~5 ng/mL)	114.1	176.46	326

Note: Purple shading indicates increased results; No shading indicates that the result is within the reference range; “\” indicates the data was missed.

coagulation function indicators shown in Fig. 2 reflected impaired coagulation function, particularly evident 17 h post-ingestion. Regarding the changes in organ function, the elevation of amylase, lipase, and four types of myocardial enzymes in Table 1 suggested damage to both pancreatic and heart functions. Additionally, certain biochemical indicators in Table 2 indicated injuries to the liver and kidney.

Pathological examinations

An autopsy was performed four days post-mortem. Upon examination of the body surface, both eyelids and bulbar conjunctiva exhibited icterus, accompanied by patchy hemorrhages. A significant area of the skin displayed hemorrhagic changes. Apart from a few surgical puncture marks, no other remarkable lesions were found.

Table 2 Representative results of tests of liver and kidney function during the hospitalization after eating mushrooms

Indicators	Times after eating mushrooms				
	12h	17h	35h	40h	59h
TBIL (0~23 $\mu\text{mol/L}$)	5.3	7.7	42.1	45.7	75.4
DBIL (0~8 $\mu\text{mol/L}$)	1.1	3.3	10.9	9.9	15.9
IBIL (1.7~21 $\mu\text{mol/L}$)	4.2	4.4	31.2	35.8	59.5
ALB (40~55 g/L)	17	23.1	38.4	34.4	30.7
GGT (10~60 U/L)	58.3	132	153	156	172
ALT (9~50 U/L)	304.5	420	562	1123	2022
AST (15~40 U/L)	587.2	779	2709	5169	10916
K (3.5~5.3 mmol/L)	4.9	3.09	3.47	\	6.95
Na (137~147 mmol/L)	165	150.8	163.8	\	151.5
Ca (2.11~2.52 mmol/L)	1.9	1.49	1.98	\	2.14
Urea (3.1~8.0 mmol/L)	9.5	10.4	10.5	\	10.3
Cr (57~97 $\mu\text{mol/L}$)	184.4	194.4	194.6	\	195.9

Note: Pink shading indicates reduced results; Purple shading indicates increased results; No shading indicates that the result is within the reference range; “\” indicates the data was missed.

Regarding the macroscopic and microscopic pathological alterations in various organs, we would present them systematically according to organ type for clarity: **Temporalis**: Extensive hemorrhages were observed in the bilateral temporal muscles, with some skeletal muscle cells undergoing necrosis and associated infiltration of mononuclear cells and lymphocytes. **Brain**: The brain weighed 1450 g and exhibited severe edema. A small subarachnoid hemorrhage was noted in the left parietal lobe, along with perivascular bleeding observed in certain areas of the parenchyma, accompanied by the presence of encephalomalacia foci (Fig. 3A). Additionally, some neurons displayed nuclear shrinkage, loss of Nissl substance, and cytoplasmic eosinophilic enhancement, indicative of ischemic-hypoxic pathological changes. There was an increase in mesenchymal glial cells and marked congestion within the small blood vessels. **Heart**: The heart weighed 350 g and presented with numerous bleeding spots across its surface, particularly on the posterior wall of the left ventricle where a patchy area of hemorrhage measuring 4 cm $\times$ 3 cm was visible. Microscopically, hemorrhagic changes were evident in both the epicardium of the left ventricle and portions of myocardial interstitium, some myocardial fibers were ruptured

alongside cellular necrosis (Fig. 3B). **Lungs**: The combined weight of both lungs was 1490 g and showed severe edema. Most alveoli demonstrated signs of hemorrhage and consolidation. Homogeneously red-stained transparent membranes adhered to alveolar walls, accompanied by significant infiltration of inflammatory cells (Fig. 3C). **Liver**: The liver weighed 1525 g and exhibited a yellow-brown appearance on both its surface and cut sections. Microscopic examination revealed disruption of hepatic cords with pronounced coagulative necrosis present in central regions of liver lobules. Many hepatocytes displayed severe fatty degeneration while there was notable accumulation of mononuclear cells and lymphocytes within portal areas (Fig. 3D). **Kidney**: The combined weight of bilateral kidneys was recorded at 325 g, several glomeruli exhibited hyaline changes alongside increased eosinophilia in proximal tubular epithelial cell cytoplasm as well as disappearance of cell nuclei—manifesting widespread coagulative necrosis (Fig. 3E). **Alimentary Tract**: Gastrointestinal bleeding was identified at multiple sites, particularly within the stomach (Fig. 3F) and small intestine (Fig. 3G), characterized by extensive bleeding coupled with cellular necrosis along with minimal inflammatory cell infiltration observed; additionally,

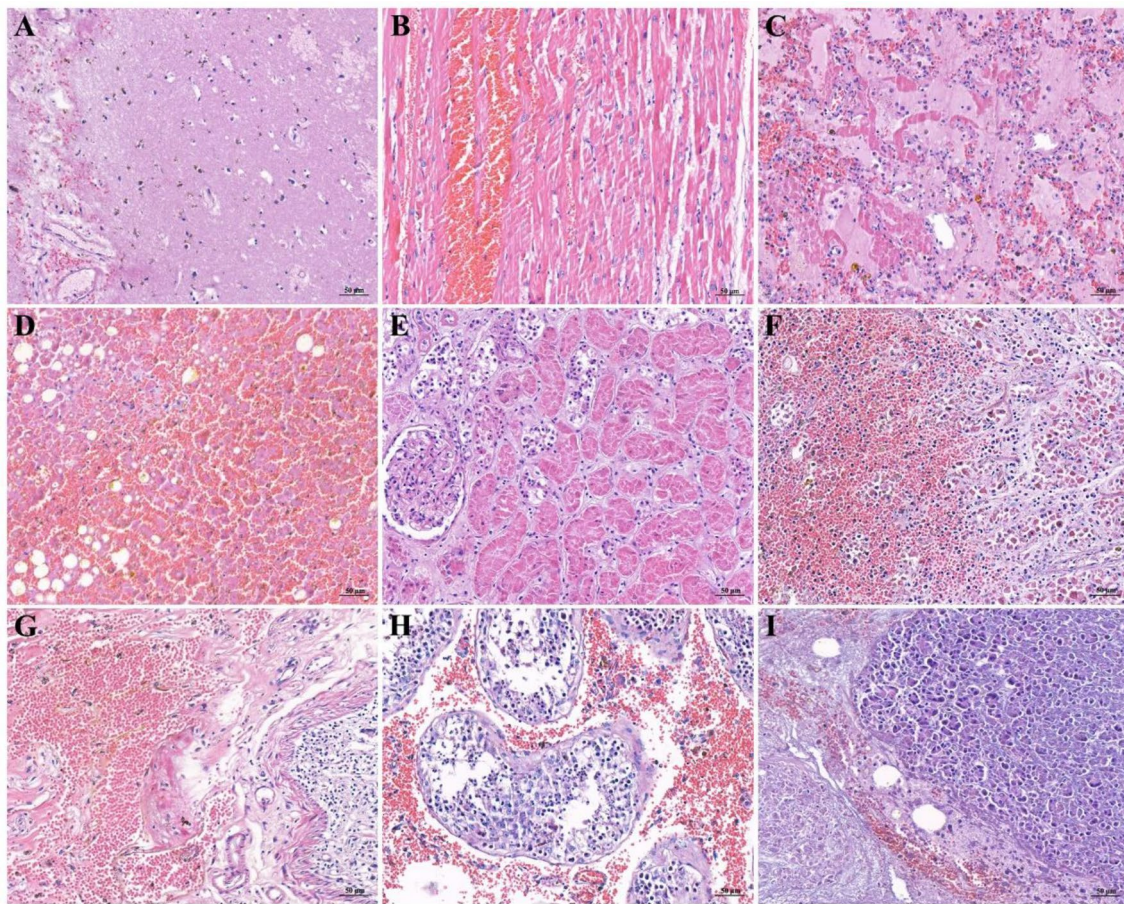


Fig. 3 Pathological changes of multiple organs in patients with mushroom poisoning. (A) Cerebral perivascular hemorrhage with softening lesion formed; (B) Myocardial interstitial hemorrhage, and muscle fiber rupture; (C) Pulmonary hemorrhage, edema, and hyaline membrane; (D) Liver fatty necrosis and coagulation; (E) Coagulated necrosis of renal

tubules; (F) Gastric mucosal hemorrhage and necrosis; (G) Lamina propria hemorrhage and necrosis of small intestine; (H) Hemorrhaging of interstitial testis, necrosis of fine ductus; (I) Pancreatic interstitial hemorrhage.)

clear thrombosis could be detected within some smaller vessels. **Testis:** Structural disorganization was noted among seminiferous tubules; significant epithelial cell necrosis occurred alongside extensive interstitial hemorrhages (Fig. 3H). **Pancreas:** Interstitial bleeding within pancreatic tissue was observed along with minor inflammatory cell infiltration present (Fig. 3I). No conspicuous abnormalities were detected in other organs.

Toxicology analysis

Ibotenic acid was successfully detected in the blood and gastric lavage fluid approximately 19 h after the consumption of mushrooms, with concentrations measured at 0.381 µg/mL and 0.202 µg/mL, respectively. Additionally, six types of common mushroom toxins (*α-amanitin*, *β-amanitin*, *γ-amanitin*, *phalloidin*, *phalloidin*, and *phalloidin*) as well as other common drugs were tested, however, the results returned negative.

Forensic diagnosis

Following a comprehensive analysis of medical records, autopsy findings, histopathological examinations, and toxicological results, we concluded that the mechanism of death was multi-organ failure resulting from mushroom poisoning.

Discussion

Mushroom toxicity represents a globally prevalent concern [5]. By the end of 2021, over 520 species of poisonous mushrooms had been identified in China alone, and this figure continues to rise [6]. Despite the extensive diversity of toxic mushrooms, fatalities caused by mushroom poisoning remain relatively infrequent. This is largely attributed to the expertise of foragers and consumers, as well as the thermal instability of most mushroom toxins; there were only 16

reported deaths among 1,303 cases of mushroom poisoning in China in 2023 [3]. Owing to the low incidence of such cases, accurate diagnosis holds both significance and challenges in clinical and forensic settings. Thus, through a comprehensive review of existing literatures and in-depth analysis of our case study, we intend to present specific clinical and pathological features that may serve as valuable references for diagnosing mushroom poisoning.

Previous researches had indicated the degree of mushroom poisoning can be preliminarily categorized into two major groups based on latency, defined as the time interval between mushroom ingestion and the onset of symptoms: (1) Latency < 6 h: functional syndromes, typically presenting with a mild course; (2) Latency > 6 h: organ-damaging syndromes, often resulting in fatal outcomes [1, 7]. However, in this case, the patient exhibited gastrointestinal symptoms, mainly abdominal pain and bloating, just half an hour after consuming the mushrooms. Subsequently, he experienced severe organ damage, which did not seem to conform to the aforementioned classification criteria. Therefore, we hypothesized that this discrepancy may be attributed to the fact that the mushrooms consumed by the patient were raw, which could have enhanced the toxicity and led to an earlier onset of symptoms.

Several reports have documented changes in clinical indicators following mushroom poisoning, reflecting damage to corresponding organs. For instance, Rahmani et al. [8] described a case of fatal mushroom poisoning where the patient succumbed to liver and kidney failure after four days, with significantly elevated levels of urea, creatinine, ALT, and AST, all of which were far beyond the reference ranges. Similar findings were observed in a retrospective analysis of mortality causes in mushroom poisoning cases [9]. Additionally, another study found that blood ammonia and lactate levels at admission were independently associated with the duration from hospitalization to death in patients with mushroom poisoning-induced liver failure. These biomarkers could potentially predict the prognosis of liver failure caused by mushroom poisoning [10]. In this report, multiple test results following mushroom ingestion indicated extensive organ damage, including the heart, liver, kidneys, and pancreas, consistent with previous studies. Furthermore, temporal variations in these test results were observed, which may be crucial for the diagnosis and management of similar mushroom poisoning cases.

The histopathological findings suggested that mushroom poisoning was the likely cause of death, as multiple organs displayed characteristic alterations. As illustrated in Fig. 3, the myocardium, lungs, liver, kidneys, and gastrointestinal tract exhibited varying degrees of hemorrhage and

necrosis. These observations were consistent with organ damage reported in other cases of mushroom poisoning [9, 11–13]. Notably, apart from the commonly affected organs, we observed circumferential hemorrhage in certain regions of the brain parenchyma, accompanied by focal encephalomalacia. This finding suggested the occurrence of toxic encephalopathy [14], a phenomenon that has been scarcely documented in the context of mushroom poisoning. Of particular interest was the discovery of extensive hemorrhage in the patient's testicular stroma, along with disorganization and necrosis of the seminiferous tubules. These findings indicated that mushroom poisoning might also cause damage to the male reproductive system, which had not been previously reported, highlighting an area that warrants further investigation.

In the identification of suspected poisoning cases, toxicological tests were crucial complements to clinical examinations and histopathological analyses. Researchers have developed several methods for the detection of mushroom toxins. Ginterová et al. [15] developed a quantitative method for three common mushroom toxins—*ibotenic acid*, *muscimol*, and *muscarine*—in urine, utilizing capillary electrophoresis coupled with electrospray tandem mass spectrometry in which the detection limits for these toxins were as low as 10 ng/mL. Bambauer et al. [16] expanded the scope to include eight different mushroom toxins, employing liquid chromatography with high-resolution mass spectrometry and the detection limit was further reduced to 1 ng/mL. Additionally, a study by another group utilized PCR analysis on undigested mushroom remnants found in stomach contents to identify mushroom species, followed by quantitative analysis using mass spectrometry [17]. These studies have significantly contributed to the identification and diagnosis of mushroom poisoning. In this case, *ibotenic acid* at concentrations of 0.381 µg/mL and 0.202 µg/mL in the patient's blood and gastric drainage fluid (preserved by the hospital) were successfully detected, respectively, approximately 19 h after mushroom ingestion by the method of liquid chromatography with high-resolution mass spectrometry.

Interestingly, *ibotenic acid* was not detected in samples taken during the autopsy, including blood, urine, liver, and stomach contents. This result was unexpected and raised several possibilities. It was speculated that the absence of *ibotenic acid* could be due to the patient's prolonged VA-ECMO and gastric lavage treatments during hospitalization, which might have facilitated the elimination of the toxin. Alternatively, *ibotenic acid* may have metabolized into unknown compounds, leading to negative test results. This finding emphasizes the importance of testing samples prior to death in similar mushroom poisoning cases, along with meticulous analysis of toxicological results.

Previous studies have confirmed that *ibotenic acid* was a neurotoxic compound and was most commonly found in two kinds of mushrooms, *Amanita muscaria* and *Amanita pantherina* [18, 19], and primarily affects the nervous system [20, 21]. However, in this case, significant damages were also observed in the circulatory, digestive, and reproductive systems. This suggested the involvement of additional, unidentified toxins that were not detected with the current analytical methods. The failure to detect other potential toxins highlights a limitation of this study, possibly due to the sensitivity or scope of our detection technology. It was conceivable that unknown substances contributed to the widespread systemic damage observed in this case.

Conclusion

This paper presented detailed temporal changes in multiple examination results during hospitalization and the pathological changes observed in various organs post-mortem in a rare case of mushroom poisoning. Moreover, we have successfully quantified the concentration of a specific mushroom toxin, aiming to provide valuable references for clinical diagnosis, treatment, and forensic determination of cause of death in similar cases.

Key points

1. A rare case of death from mushroom poisoning was reported;
2. Temporal changes in clinical indicators and pathological changes were shown in detail;
3. A kind of mushroom toxin (ibotenic acid) was successfully detected qualitatively and quantitatively.

Supplementary Information The online version contains supplementary material available at <https://doi.org/10.1007/s12024-025-00956-2>.

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Declarations

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Consent for publication The consent for publication had been obtained from Huazhong University of Science and Technology.

Informed consent Informed consent from the family of the deceased.

Competing interests There was no competing interest.

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